

PREETHI AMARANENI

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SUMMARY:

Business/Data Analyst with 4+ years in financial services, delivering analytical and reporting solutions through SQL, Excel, Power BI, and Tableau. Experienced in building automated data pipelines using Python, SQL, and AWS to improve reporting efficiency and data quality. Skilled in statistical modeling with R, SAS, and Python, and translating complex business requirements into actionable insights.

PROFESSIONAL EXPERIENCE:

Data Analyst | Capital One - Charlotte, NC

January 2025 - Present

- Built Python and SQL-driven data pipelines to automate aggregation of customer product usage and marketing engagement data into Redshift, reducing manual reporting delays and supporting near real-time insights for the product analytics team.
- Developed Tableau dashboards to track product activation funnels, dealership sales patterns, and campaign response rates, enabling faster iteration on go-to-market strategies by cross-functional marketing teams.
- Conducted A/B test analysis on personalized messaging strategies using Python (pandas, statsmodels) and presented statistically validated insights to product teams, directly influencing customer segmentation approaches for Q2.
- Configured AWS S3 buckets and Lambda functions to ingest campaign performance files from third-party vendors, streamlining ingestion workflows and enhancing data quality monitoring.

Data Analyst | TIAA Financial Services - Charlotte, NC

August 2023 - October 2024

- Developed scalable business reporting packages in Excel (pivot tables, Power Pivot) and SQL for retirement accounts, delivering consistent metrics to senior stakeholders and improving turnaround time for ad hoc reports by 25%.
- Built Power BI and Tableau dashboards to visualize key retirement account KPIs, fraud alerts, and compliance trends, enabling data-driven decisions and improving cross-team transparency.
- Automated recurring Excel-based reports using VBA macros, reducing manual effort in data formatting and monthly fund performance distribution by over 40%.
- Integrated automated anomaly detection workflows using Alteryx and SQL, reducing investigation time by 25% and supporting vendor-facing fraud monitoring capabilities.
- Constructed robust ETL pipelines using AWS services (S3, Lambda, Redshift), supporting large-scale data ingestion and improving reporting freshness for plan sponsors by 20%.
- Designed a Snowflake-based reporting layer to centralize curated financial datasets for plan sponsor reporting, implemented RBAC for data security, and optimized performance using clustering keys integrated with Power BI for faster and more reliable KPI dashboards.
- Documented end-to-end data lineage for retirement account, compliance, and fraud monitoring datasets, mapping data flow from source ingestion to final reporting in Power BI/Tableau.
- Designed exception tracking logic in SQL to flag outliers in fund movements and data mismatches, aiding in early detection of account discrepancies and reducing downstream errors.
- Collaborated with retirement operations teams to analyze account servicing delays and modeled turnaround times using Python, identifying process gaps that led to a 15% improvement in SLA compliance.
- Conducted regression analysis in Python to evaluate the impact of plan design changes on fund movement volumes, helping guide product strategy.
- Conducted A/B testing on communication strategies for plan sponsors to evaluate the effectiveness of different messaging formats; results showed a 17% lift in broker engagement with customized insights versus generic updates.

Data Analyst | Fidelity Investments - Charlotte, NC

May 2022 - May 2023

- Designed and deployed scalable data ingestion pipelines using AWS Lambda and DMS to move capital markets and client portfolio data into Redshift, reducing data refresh cycles by 35%.
- Developed reusable VBA macros to streamline complex workbook operations, including reconciliation flagging, formatting automation, and scheduled data rollups for portfolio audits.
- Wrote complex SQL queries on Redshift and SQL Server to analyze transaction-level investment data, enabling real-time insights into trade behavior and SLA adherence.
- Created interactive Power BI dashboards visualizing capital market activity, high-risk trade patterns, and settlement delays, supporting proactive decisions by investment and compliance teams.

- Defined KPIs with product and finance teams and implemented a standardized data model in Snowflake and Redshift, ensuring consistency across self-service reporting tools.
- Maintained comprehensive data lineage documentation for capital markets and portfolio datasets by capturing table-level and field-level transformations in Redshift/Snowflake. Provided stakeholders with clear traceability for regulatory reporting and accelerated impact assessments for schema changes
- Configured AWS CloudTrail and IAM roles to enable secure access logging and enforce data governance policies across analytical datasets.
- Integrated version control and deployment tracking using Git and Jenkins, supporting CI/CD pipelines for dashboard rollouts and stored procedure updates.
- Conducted time series forecasting and regression modeling in R and Python to evaluate trading volatility and seasonal behavior, improving pre-trade compliance triggers.
- Designed and ran A/B tests to assess the impact of dashboard layout changes on portfolio manager usage rates; variant B with dynamic filters resulted in 22% higher interaction rates and faster SLA exception identification.
- Participated in the design of data vault models within Redshift and Snowflake to support long-term archival and reporting of high-volume financial transactions.
- Delivered Tableau-based exception dashboards using AWS-hosted data, empowering operations teams to monitor trade errors and reduce escalations.

Data Analyst | TATA CAPITAL LIMITED - India

June 2020 - July 2021

- Developed ETL pipelines using Python and SQL to consolidate transactional data into AWS Redshift and MS SQL Server, improving data availability for reporting.
- Built Excel-based reporting tools using pivot tables, nested IF statements, and array formulas to analyze loan performance, customer segmentation, and branch-level metrics.
- Created reusable SQL queries to extract, clean, and aggregate lending data for monthly executive summaries and risk assessment reports.
- Developed interactive Tableau and Power BI dashboards to monitor KPI performance across business units, including disbursement rates, NPA ratios, and SLA adherence.
- Combined JSON and log data using PySpark and flattened them for dashboard integration, helping uncover patterns in customer support and loan repayment behaviors.
- Automated repetitive Excel-based workflows using macros and Power Query, saving ~15 hours/week of manual effort and improving consistency in report delivery.
- Supported data validation and accuracy checks across SQL Server and Redshift environments before final dashboards and reports were released to senior management.
- Modeled NPA trends and loan aging using Python's sklearn, enabling leadership to predict high-risk segments and adjust credit strategies.
- Assisted in designing a Snowflake schema for post-loan disbursement analysis, enhancing reporting granularity for portfolio review meetings.

TECHNICAL SKILLS:

Languages & Scripting: Python (Pandas, NumPy, PySpark), SQL, PL/SQL, Shell (Unix/Linux)

ETL & Workflow Orchestration: Azure Data Factory, Azure Functions, Azure DevOps, AWS Lambda, AWS DMS, Informatica, Alteryx, Jenkins, Apache Airflow

Cloud Platforms: Azure (ADF, Synapse, Key Vault, DevOps), AWS (S3, Redshift, Lambda, DMS, CloudTrail), Snowflake

Big Data & Streaming: Apache Spark, Hadoop, Hive, Databricks, PySpark

Databases: SQL Server, Oracle, PostgreSQL, DB2, Redshift, Teradata, MongoDB, DynamoDB

Visualization & Analytics: Power BI, Tableau, Excel (Advanced Formulas, Power Query, Pivot Tables, VLOOKUP), Power Apps

DevOps & Collaboration: Git, GitHub Actions, Azure DevOps, Jupyter Notebooks, Jira

EDUCATION:

Trine University | Master of Science - Information Science

K L University | Bachelor of Technology - Electronics & Communication Engineering

August 2021 | GPA: 3.9/4.0

May 2021 | GPA: 3.75/4.0